Car Cost Analysis Interpretation

Revised graph results using the either-car edge rule

Purpose and scope

This report interprets the supplied 10-row car dataset, its linear cost projection, the cost visualizations, and the revised graph model. Each car record is treated as its own observation and graph vertex, labeled by its recorded cost. The graph interpretation in this report is that a pair receives an edge when either car meets the relevant speed or seats condition.

The graph metrics are exact for this graph definition and the supplied values. The future cost figures are a mathematical extrapolation from a small sample and should not be treated as validated market-price predictions.

Cost trend and 10-year projection

An ordinary least-squares line was fitted with year as the predictor and cost as the response. For these records, the equation is cost = −20,160,000 + 10,000 × year. The slope means the fitted cost rises by 10,000 for each one-year increase in the supplied data. The model has an in-sample R^2 of 1.00 because all 10 rows lie exactly on this straight line.

Continuing that line produces the following projections. These are model outputs, not independently verified future prices:

Year	Projected cost
2033	170,000
2034	180,000
2035	190,000
2036	200,000
2037	210,000
2038	220,000
2039	230,000
2040	240,000
2041	250,000
2042	260,000

The exact linear pattern and perfect in-sample fit do not establish that real car values will keep rising at this rate. The model uses year alone and has no independent test period. It omits factors such as mileage, make

and model, vehicle condition, inflation, and changing market conditions. R^2 describes fit to these observed rows; it does not measure future predictive accuracy. Treat the 2033–2042 values as an illustrative continuation of the sample pattern.

Cost visualizations

Scatterplot

Each point represents one car's year and recorded cost. The fitted line summarizes the observed linear pattern, and its extension after the latest observed year shows the decade projection. A point's vertical distance from the fitted line is its residual.

Boxplot

The box shows the middle half of the observed costs, the internal line marks the median, and the whiskers summarize the range under the plotting convention. Potential outliers are prompts to inspect records, not proof of error. This plot describes the 10 supplied cars only.

Histogram

The histogram groups the observed costs into ranges and shows how many records fall in each range. Its appearance depends partly on the selected bin widths.

Frequency polygon

The frequency polygon connects the histogram frequencies at the bin midpoints. It is another view of the same grouped cost observations, not a separate dataset or independent result.

Revised graph definition

Each car record is a vertex labeled with its cost. A pair receives a green speed-rule relation if at least one of the two cars has a speed from 500 through 700 km/hr, inclusive. A pair receives a yellow seats-rule relation if at least one of the two cars has at least 3 seats. The graph is the union of these relations; if a pair meets both conditions, both colored relations are drawn. Vertices are colored blue for display.

Because all edges join pairs of vertices, the structure is an ordinary undirected graph, or a 2-uniform hypergraph. A general hypergraph may have edges containing three or more vertices; this pairwise rule does not create those larger edges.

Graph structure and metrics

Nine cars in the dataset have at least 3 seats, and one car has 2 seats. Under the either-car seats rule, any pair contains at least one of the nine qualifying cars: a pair of qualifying cars meets the rule at both endpoints, and a pair containing the single 2-seat car has a qualifying car at the other endpoint. Therefore every pair is connected by a yellow seats-rule edge. The union graph is the complete graph K_{10} with 10 vertices and 45 edges. The green speed relations are additional colored relations on some of those already-connected pairs; they do not add new pairs to the union graph.

ic	Value	Interpretation
ces	10	One per car record.
n edges	45	Every pair of the 10 vertices is connected; the graph is K_{10}.
matic number χ(G)	10	Every vertex is adjacent to every other vertex, so each needs a different vertex color.
matic index χ'(G)	9	Minimum edge colors so edges sharing a vertex differ. For the even complete g K_{10}, this is 9.
pendence number α(G)	1	No two distinct vertices are nonadjacent, so the largest independent set has o vertex.
sing number cr(G)	60	Minimum crossings over all drawings of K_{10}; this is a graph invariant, not the crossings in one plotted layout.
ue number ω(G)	10	All 10 vertices form a clique because every pair is adjacent.

How to interpret the graph measures

The blue vertex fill is a display choice. The chromatic number is a separate mathematical measure of how many colors would be required to color adjacent vertices differently, so it is 10 even though the plotted nodes are all blue. Green and yellow indicate the two edge criteria; they are not the edge-coloring solution measured by the chromatic index.

The crossing number is the minimum possible number of crossings over every drawing of the graph. A particular spring-layout plot can show more crossings than this minimum; its visible crossings do not change the reported value of 60.

Overall conclusion

For the supplied data, cost increases by exactly 10,000 per year, producing a perfect in-sample linear fit and a projection from 170,000 in 2033 to 260,000 in 2042. The projection is illustrative because it extends a simple pattern from only 10 rows. Under the revised either-car graph rule, every pair is adjacent through the seats condition, making the graph K_{10}. Its reported graph metrics follow from that complete-graph structure.